

Sequential Types as a Universal Basis for CRDT Design

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Abstract

Placeholder text

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1 Introduction

2 Background

3 CRDT Liftability

Definition 3.1 (Sequential Type). A sequential type $T = (S, s_0, \Delta)$ is a triple where S is the set of states of an object of type T , $s_0 \in S$ the initial state and $\Delta : S \times Op \rightarrow S$ is the transition function mapping a state and an operation to a successor state.

We write $s \xrightarrow{op} s'$ when $\Delta(s, op) = s'$

Definition 3.2 (Induced binary operator). Given a sequential type T , we define the binary operator $^1 \oplus : S \times S \rightarrow S$ induced by the transition function Δ such that for any states $s, s', s'' \in S$, $s \oplus s' = s''$ by applying the union of the operation multi-sets that lead to states s and s' respectively.

Formally, if $s_0 \xrightarrow{op_1 \dots op_m} s$ and $s_0 \xrightarrow{op'_1 \dots op'_n} s'$. Then, $s \oplus s' = \Delta^*(s_0, \{op_1 \dots op_m\} \uplus \{op'_1 \dots op'_n\})$

Definition 3.3 (Liftability). A sequential type T admits a $^\Delta$ -lifting if there exists a CRDT type $T^\Delta = (\mathbb{S}, \perp, \Delta')$ with a binary operator $\sqcup$, an observation function $get : \mathbb{S} \rightarrow S$ that materializes the value of a CRDT and satisfies the following:

(L1) $get(\perp) = s_0$

(L2) For any $s_1, s_2 \in \mathbb{S}$, $get(s_1 \sqcup s_2) = get(s_1) \oplus get(s_2)$

(L3) get is surjective: for every $s \in S$ there exists some $s' \in \mathbb{S}$ with $get(s') = s$.

(L4) $\sqcup$ is commutative, associative, and idempotent.

¹The binary operator $\oplus$ is well-defined only when the transition system is confluent.

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Theorem 3.4 (Liftability Characterization). A sequential type T is liftable via $^\Delta$ if T is a commutative monoid.

Proof. We prove both directions separately. ($\Rightarrow$) **Necessity: Liftability implies commutative monoid**

We verify the three axioms of a commutative monoid in turn.

Associativity of $\oplus$. Let $s, t, u \in S$ be arbitrary. By surjectivity (L3), choose $a, b, c \in \mathbb{S}$ with $get(a) = s$, $get(b) = t$, $get(c) = u$.

By (L2) and associativity of $\sqcup$ (L4):

$$get(a \sqcup (b \sqcup c)) = get(a) \oplus get(b \sqcup c) = s \oplus (t \oplus u) \quad (1)$$

$$get((a \sqcup b) \sqcup c) = get(a \sqcup b) \oplus get(c) = (s \oplus t) \oplus u \quad (2)$$

Since $\sqcup$ is associative, $a \sqcup (b \sqcup c) = (a \sqcup b) \sqcup c$, and applying get to both sides:

$$s \oplus (t \oplus u) = (s \oplus t) \oplus u$$

Since s, t, u were arbitrary, $\oplus$ is associative. $\blacktriangleright$ (Pierre:) OK $\blacktriangleleft$

Identity element. Let $t \in S$ be arbitrary and choose $a \in \mathbb{S}$ with $get(a) = t$. We need a state representing s_0 to merge with. By (L1), $get(\perp) = s_0$. We need $\perp$ to act as a fresh contribution with respect to a , which holds at initialization since $\perp$ carries no updates.

By (L2):

$$get(a \sqcup \perp) = get(a) \oplus get(\perp) = t \oplus s_0$$

But $\perp$ is the identity of $\sqcup$ by the CRDT definition, so $a \sqcup \perp = a$, hence:

$$get(a \sqcup \perp) = get(a) = t$$

Therefore $t \oplus s_0 = t$. By a symmetric argument using $\perp \sqcup a = a$:

$$get(\perp \sqcup a) = get(\perp) \oplus get(a) = s_0 \oplus t = t$$

So s_0 is a two-sided identity for $\oplus$. $\blacktriangleright$ (Pierre:) OK $\blacktriangleleft$

Commutativity of $\oplus$. Let $s, t \in S$ be arbitrary. By surjectivity (L3), choose fresh $a, b \in \mathbb{S}$ with $get(a) = s$ and $get(b) = t$.

By (L2):

$$get(a \sqcup b) = get(a) \oplus get(b) = s \oplus t$$

By commutativity of $\sqcup$ (L4):

$$a \sqcup b = b \sqcup a$$

Applying get and (L2) to the right hand side:

$$get(b \sqcup a) = get(b) \oplus get(a) = t \oplus s$$

Since $get(a \sqcup b) = get(b \sqcup a)$, we conclude:

$$s \oplus t = t \oplus s$$

Since s, t were arbitrary, $\oplus$ is commutative.

This completes the proof of necessity. $\blacktriangleright$ (Pierre:) OK $\blacktriangleleft$ $\square$

($\Rightarrow$) Sufficiency: Commutative monoid implies liftability

Assume that T is a commutative monoid. We construct a valid Δ -lifting explicitly.

Construction: We define $T^\Delta = (\mathbb{S}, \sqcup, \Delta')$ as follows:

- **Initial element:** $\perp := \lambda r. s_0$, the constant function returning s_0 .
- **Merge operator:** $(f \sqcup g)(r) := f(r) \oplus g(r)$, applied pointwise.
- **Observation function:** $get(f) := \bigoplus_{r \in R} f(r)$, the fold of $\oplus$ over all replica values.

Construction. Let R be a finite set of replica identifiers. ▶(Pierre:) One interesting (but not directly related) question is how necessary this assumption is. For instance, what happens is the system is anonymous. This can be useful to bound metadata usage. ▶(Ayush:) The replica identifiers here, simplify the get function w.r.t freshness constraints. I will try to find a way to eliminate this. ▶

Define:

$$T^\Delta := (R \rightarrow T)$$

that is, the type of functions assigning a value in T to each replica, which we think of as a state vector. Define:

- **Initial element:** $\perp := \lambda r. e$, the constant function returning e .
- **Merge operator:** $(f \sqcup g)(r) := f(r) \oplus g(r)$, applied pointwise.
- **Observation function:** $get(f) := \bigoplus_{r \in R} f(r)$, the fold of $\oplus$ over all replica values.

Each replica r maintains only its own coordinate, so replica r 's update is:

$$\delta_r(f, t)(r') := \begin{cases} f(r') \oplus t & \text{if } r' = r \\ f(r') & \text{otherwise} \end{cases}$$

▶(Pierre:) OK. ▶

Verification of CRDT axioms.

Commutativity of $\sqcup$: For any f, g and any r :

$$(f \sqcup g)(r) = f(r) \oplus g(r) = g(r) \oplus f(r) = (g \sqcup f)(r)$$

using commutativity of $\oplus$. Hence $f \sqcup g = g \sqcup f$.

▶(Pierre:) We might need also that e is neutral; no? This is easy: $r, (f \sqcup \perp)(r) = f(r) \oplus \perp(r) = f(r) \oplus e = f(r)$. ▶(Ayush:) The proof of e (now called s_0) being neutral comes from the other direction where we define the CRDT type and prove that it produces a commutative monoid. ▶

Associativity of $\sqcup$: Follows immediately from associativity of $\oplus$ applied pointwise.

Idempotency of $\sqcup$: This requires an additional condition. Since $\oplus$ need not be idempotent in general, we enforce idempotency structurally by the freshness discipline. Each replica update inc_r is applied at most once per value, so the case $f \sqcup f$ never arises in practice. Alternatively, one can replace T with the lifted multiset construction $T^\Delta := (R \rightarrow \mathbb{N}) \otimes T$ and

recover idempotency formally, at the cost of a more complex get. ▶(Pierre:) Not sure to follow this last part. ▶

Verification of the homomorphism condition (L2). Let $f, g \in T^\Delta$ be fresh with respect to one another, meaning their nonzero coordinates are disjoint: for each r , at most one of $f(r), g(r)$ is non-initial. ▶(Pierre:) I see. ▶ Then:

$$get(f \sqcup g) = \bigoplus_{r \in R} (f \sqcup g)(r) = \bigoplus_{r \in R} (f(r) \oplus g(r))$$

Since $\oplus$ is commutative and associative, and the supports of f and g are disjoint:

$$= \left(\bigoplus_{r \in R} f(r) \right) \oplus \left(\bigoplus_{r \in R} g(r) \right) = get(f) \oplus get(g)$$

Verification of (L1).

$$get(\perp) = \bigoplus_{r \in R} e = e$$

using the identity law of $\oplus$ iterated over R .

Verification of (L3). For any $t \in T$, fix any replica $r_0 \in R$ and define $f := \lambda r. \text{if } r = r_0 \text{ then } t \text{ else } e$. Then $get(f) = t \oplus e \oplus \dots \oplus e = t$. So get is surjective. ▶(Pierre:) OK. ▶

This completes the construction and verification. Hence $(T, \oplus, e)$ admits a Δ -lifting. □

Conclusion

$(T, \oplus, e)$ admits a Δ -lifting if and only if $(T, \oplus, e)$ is a commutative monoid. ■

Corollary 3.5 (Non-liftability of string concatenation). *The type $(\Sigma^*, \cdot, \epsilon)$ admits no Δ -lifting, since string concatenation is not commutative: “AB” · “CD” = “ABCD” ≠ “CDAB” = “CD” · “AB”. By the necessity direction of the theorem, no valid get can exist.* □

4 Extended Classification of Unliftable Types

4.1 Types with No Binary Operation At All

▶(Pierre:) See my point above about inducing it from the transition function. ▶

Consider a type T that carries no natural merge structure whatsoever. The canonical example is:

$$T = \{0, 1\}^* / \sim$$

the set of bitstrings modulo some equivalence that is not a congruence with respect to any commutative operation. More concretely:

Corollary 4.1 (Last-Write-Wins Register). *Let T be any set, with the intended sequential semantics being that the most recent write wins. The “operation” is:*

$$a \oplus b = b \quad (\text{take the rightmost value})$$

This is not commutative since $a \oplus b = b \neq a = b \oplus a$ in general. Therefore no $\blacktriangle$ -lifting exists.

However, this is not because the type is unliftable in principle. It is because the sequential specification itself is inherently order-dependent, and no CRDT can implement it without additional metadata. The standard resolution is to enrich T with timestamps, yielding a different sequential type $(T \times \text{Timestamp}, \oplus_{\text{LWW}}, \perp)$ that is liftable. The original T is unliftable but only because it is under-specified. $\square$

4.2 Irreparably Unliftable Types

The deepest obstruction arises from types whose semantics are inherently non-monotone. A function $f : T \rightarrow T$ is monotone with respect to a partial order if:

$$a \leq b \Rightarrow f(a) \leq f(b)$$

CRDTs are monotone by construction — state only moves upward in the semilattice order. Therefore:

Theorem 4.2 (Monotonicity Obstruction). *If the sequential specification of $(T, \oplus, e)$ requires any operation that is non-monotone with respect to every partial order on T , then T admits no $\blacktriangle$ -lifting.*

The canonical example is:

$$T = (\mathbb{Z}, \min, +\infty)$$

considered not as an output type but as a type supporting decrement without bound. A counter that can decrease arbitrarily has no monotone CRDT representation, because any

decrease would require moving downward in the lattice order, contradicting the SEC requirement that state transitions are monotone.

More precisely:

Theorem 4.3 (Non-liftability of unbounded decrement). *Let $T = (\mathbb{Z}, +, 0)$ with the operation set including both increment and decrement as primitive sequential operations rather than as derived from a PN-Counter lifting. Then T admits no $\blacktriangle$ -lifting that is monotone and satisfies SEC.*

Proof. Suppose $T^\blacktriangle$ exists. Since SEC requires that the state lattice is a join-semilattice, the state space must have a partial order $\leq$ such that every update is monotone increasing. But the decrement operation $d : T^\blacktriangle \rightarrow T^\blacktriangle$ must satisfy:

$$\text{get}(d(a)) = \text{get}(a) - 1 < \text{get}(a)$$

So d strictly decreases the observable value. For d to be monotone, we need $d(a) \geq a$ in the lattice order, yet get must be order-preserving, so $\text{get}(d(a)) \geq \text{get}(a)$, contradicting $\text{get}(d(a)) = \text{get}(a) - 1$. Therefore no such lifting exists. $\square$

Remark. The PN-Counter resolves this not by lifting $(\mathbb{Z}, +, 0)$ directly, but by changing the type to a pair of monotone counters $(\mathbb{N} \times \mathbb{N}, \oplus_{\text{PN}}, (0, 0))$ whose get computes the difference. The integer type itself remains unliftable.

Type class	Liftable?	Reason
Commutative monoids ($\mathbb{N}, +, 0$), ($\mathbb{B}, \vee, \perp$)	✓ Yes	Constructive lifting exists (Theorem 1)
Non-commutative monoids ($\Sigma^*, \cdot, \epsilon$)	× No	Commutativity of $\sqcup$ forces commutativity of $\oplus$
Non-associative types ($\mathbb{Z}, -, 0$)	× No	Associativity of $\sqcup$ forces associativity of $\oplus$
Non-monotone types ($\mathbb{Z}, +, 0$) with decrement	× No	SEC monotonicity is violated
Under-specified types (LWW register)	× No directly ✓ with enrichment	Liftable only after adding metadata to the type

Table 1. classification of types with respect to liftability